

Tracks and Suspension

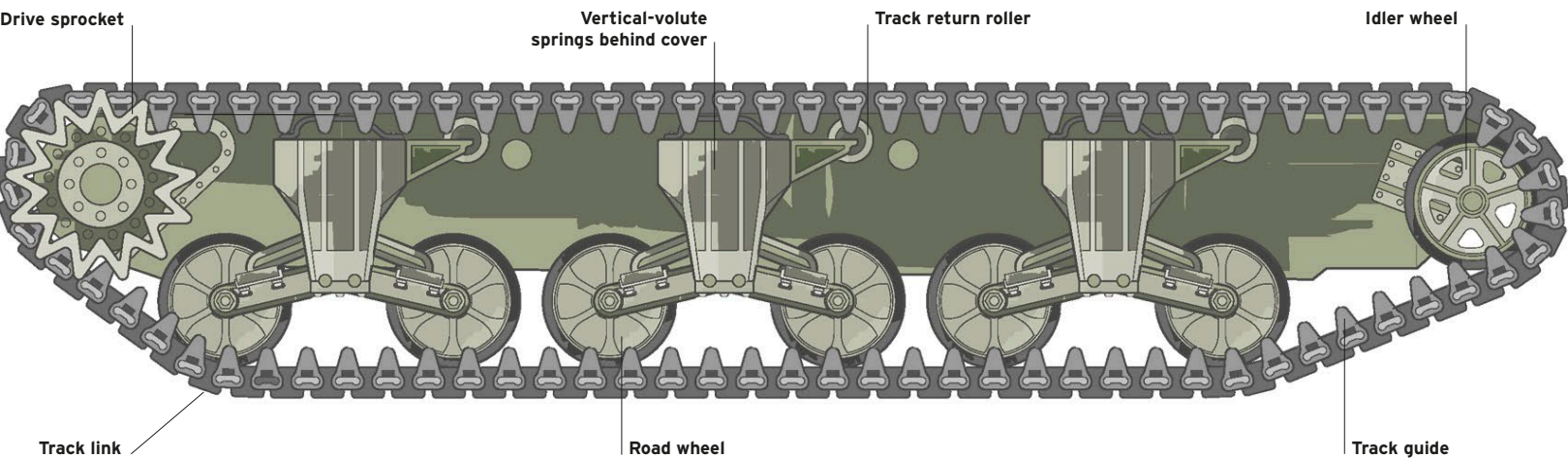
British tanks of World War I had no sprung suspension system at all: the tracks simply ran over fixed rollers. As a result, the ride was nothing short of chaotic, and crew members risked serious injury. The French Schneider and St. Chamond used simple leaf and coiled spring systems, which were only marginally better, although the light FT-17 improved on the basic principle. J. Walter Christie's original hybrid system, as demonstrated in 1919, was a real step forward, as was the leaf-

spring system adopted for the Vickers Medium in 1922. However, it was not until Christie unveiled his M1928, with lengthened suspension travel, that top speed increased dramatically—even if it was rejected by the armed forces of his native US, and only adopted by the UK and the Soviet Union. In the meantime, more complex Horstmann and volute-spring systems became popular, but both eventually gave way to much simpler, and cheaper, torsion bars.

CONTINUOUS TRACK

It was accepted from the outset that the most reliable way of moving a heavy armored vehicle across the battlefield was by way of “continuous” tracks, even though the system had some drawbacks, including high cost, low durability, and the vulnerability of the entire vehicle if a single track segment was damaged. The design of the tracks themselves and the way in which links

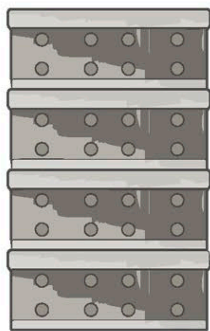
were joined was a matter of concern, too, as were factors such as whether they should be driven from the rear or from the front—which determined whether the upper, “return,” track or the load-bearing lower one should be under tension; each had their pros and cons. Finally, there was the issue of where the tracks should be located and how they should be held in place.



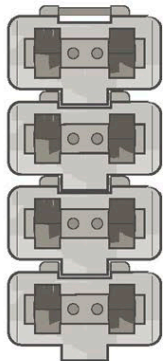
TYPES OF TRACK

The earliest continuous tracks were simple strips of metal that were connected by hinges to form a closed loop. They were unable to move sideways, and thus were easily shed, and were prone to

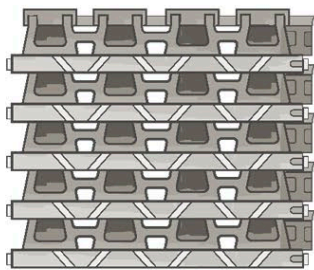
slippage. It was more than a decade before designs evolved that enabled lateral movement by means of track guides and grips that provided adequate traction on both hard and soft ground.



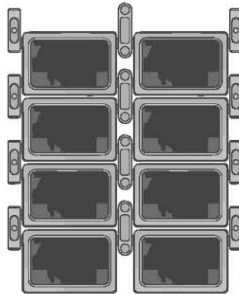
Tank Mark IV
The earliest tracks had link-wide hinges and shallow flanges for grip.



Vickers Medium
The Medium had narrow links with short hinges that provided flexibility.



PzKpfw IV Tiger
The Tiger had wide, aggressive tracks for combat and narrower ones for transportation.



M1 Abrams
Like many modern tanks, the Abrams' tracks have removable rubber pads.

DRIVE SPROCKETS

Although they started out as simple toothed wheels, drive sprockets evolved into much more complex assemblies over the years, incorporating reduction gears and a free-wheeling capability. They are mounted at the rear of most modern tanks, putting the lower track run under tension, which reduces wear on all major components.

